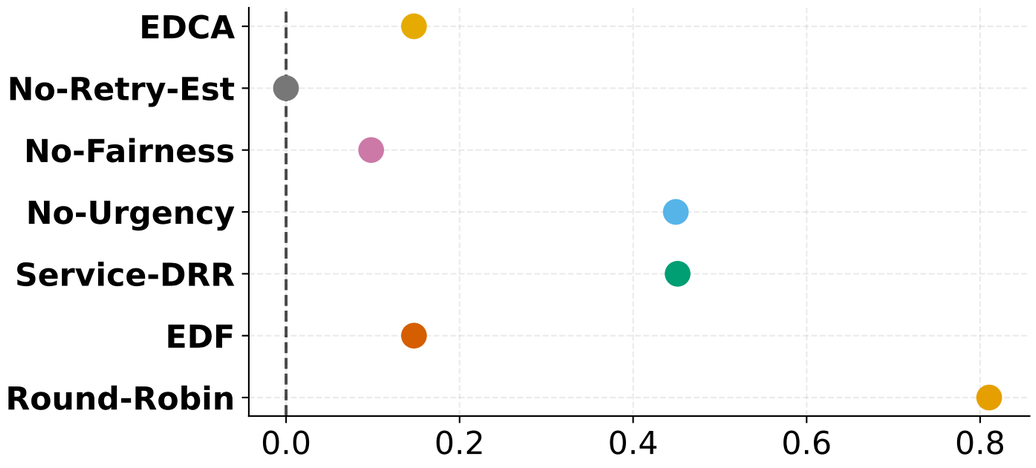
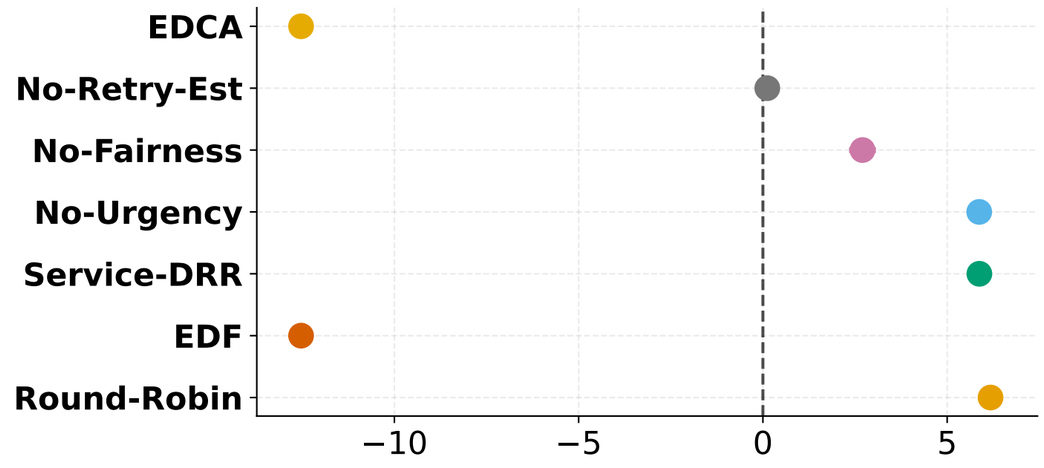
# Closed-Loop TCP Reno Paired Improvements

## Positive improvement favors U-AFS • Stars (*): Holm-significant • Bars: 95% bootstrap CIs

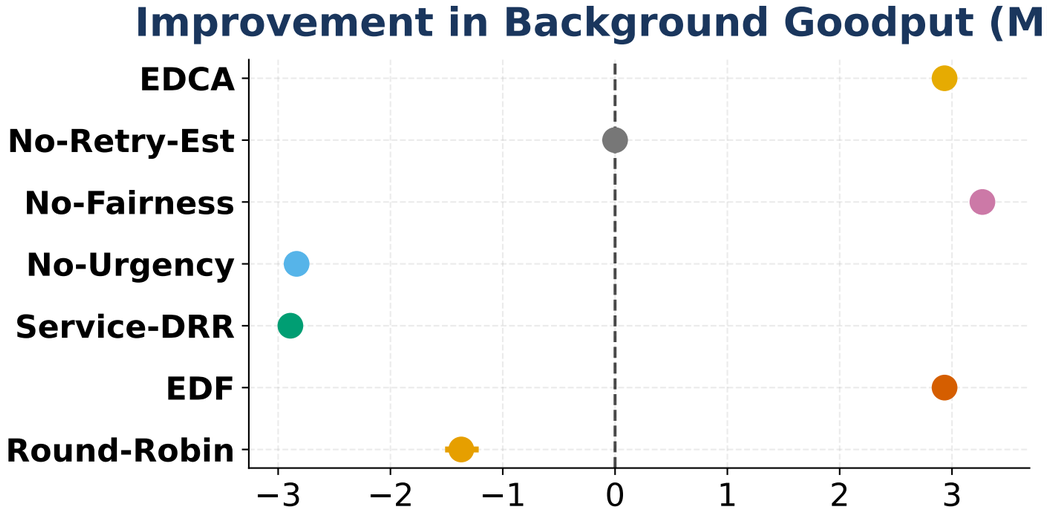
### Improvement in Real-Time Miss Fraction



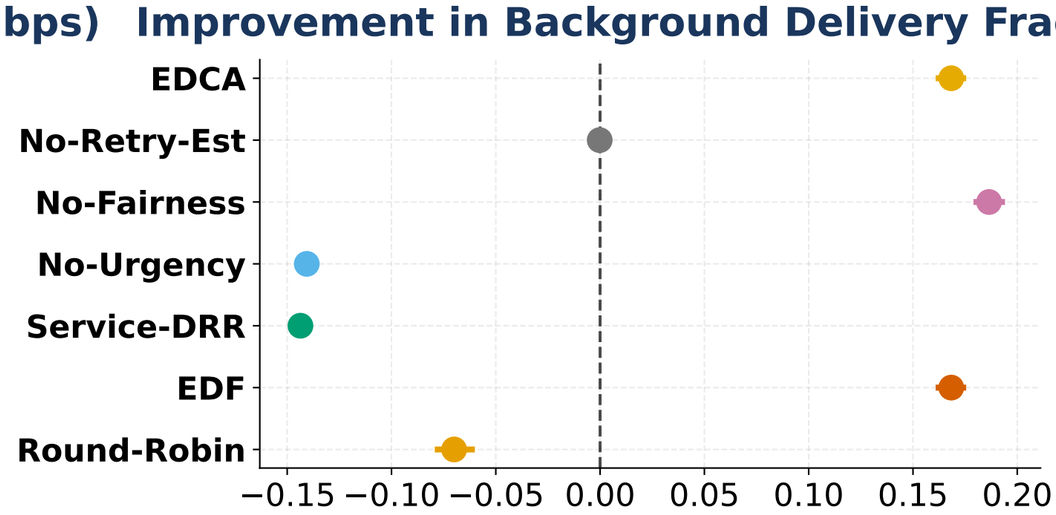

### Improvement in RT 99th-Percentile Delay (ms)



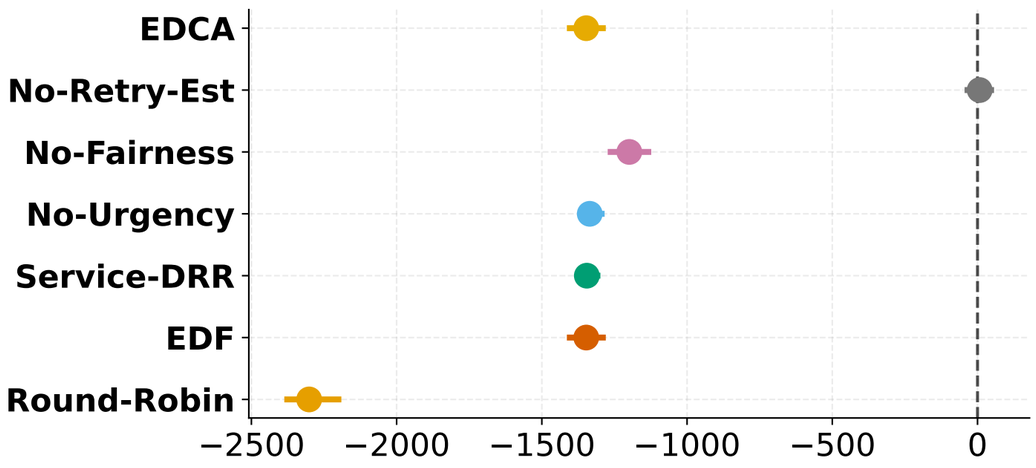

### Improvement in Background Goodput (Mbps)



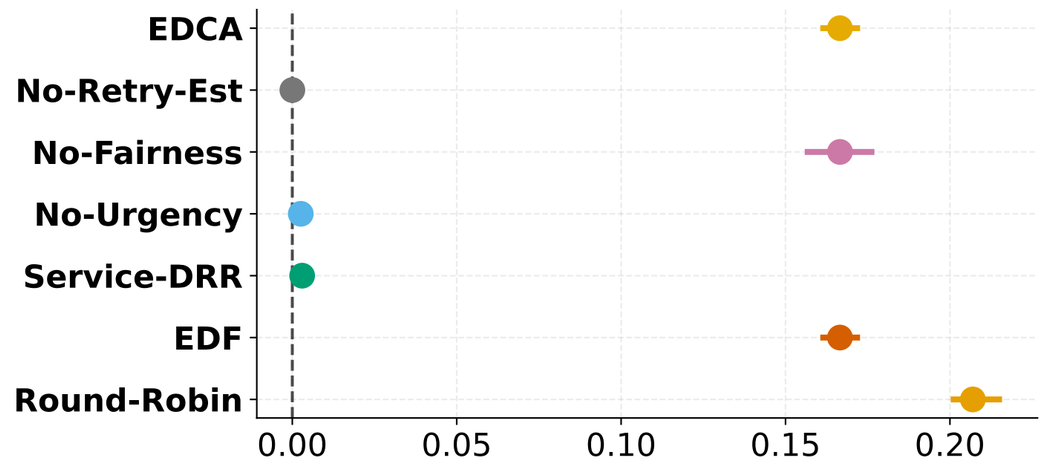

### Improvement in Background Delivery Fraction



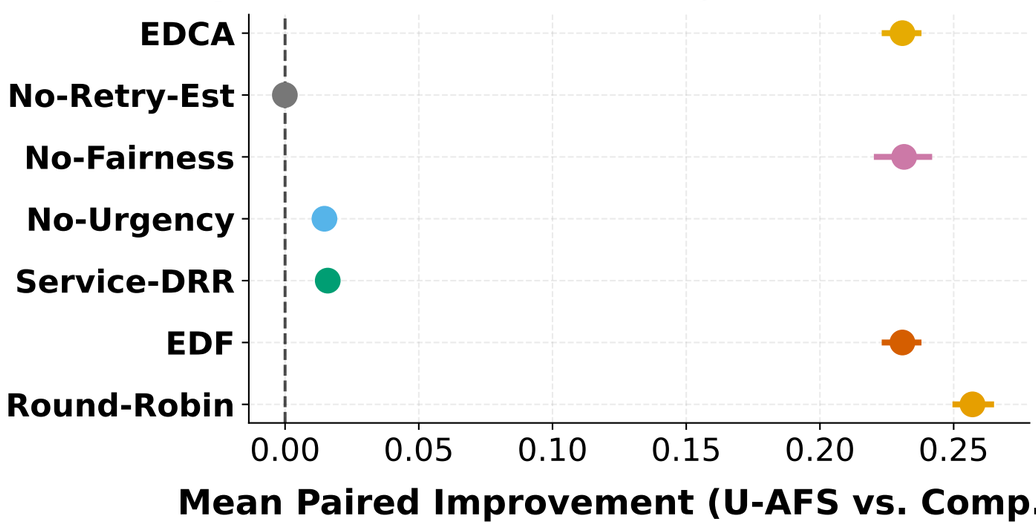

### Improvement in Background p95 Delay (ms)




### Improvement in Service Jain Fairness Index




### Improvement in Airtime Jain Fairness Index




STATISTICAL METHODOLOGY GUIDE

- Positive improvement indicates U-AFS outperforms comparator.
- Star (*): Statistically significant under two-sided paired Wilcoxon signed-rank test with global Holm-Bonferroni family-wise error rate control ($p < 0.05$).
- Circle (o): Non-significant difference.
- Horizontal error bars: Unadjusted 95% paired bootstrap CIs.